Table: Comparative analysis of literatures in SRR-AFSC

S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
01	Stanescu et al. (2025)	Technology roles in enhancing the sustainability of AFSC	develop a comprehensive and systematic analysis of the application of blockchain technologies and related digital solutions in AFSC	AFSC	literature review, conceptual approach	convergent integration of emerging digital technologies represent a strategic direction for strengthening resilience, transparency, and sustainability by ensuring traceability, fraud prevention, adoption factors, emerging innovations and applications	PRISMA	lack of empirical validation	NA	NA
02	Amamou et al. (2025)	sustainability themes	develop a multi-capital sustainability framework	AFSC	literature review	116 indicators are identified	PRISMA	Lack for empirical validation	NA	NA
03	Dieguez-Santana et al. (2025)	Applications of the circular economy	identify new trends and gaps	circular AFSC	literature review	the clusters are innovation and sustainability in AFSCs, barriers to sustainable management, strategies to reduce food loss and waste, resource recovery and food security, and conceptual frameworks for waste reduction	bibliometric analysis, SLR, PRISMA	time scale	NA	NA
04	Latino et al. (2025)	role of Iot and data analytics in decision-making	provide a roadmap for companies to integrate technology for sustainable operations and trust consumer communication	circular AFSC	multi cases approach	data integration facilitates sustainable decision-making in AFSC and facilitate transparent consumer communication	literature review, depth interview, analytic design	sample	tomatoes, seafood	Italy
05	Reitano, Segovia, and Nayga (2025)	impact of AI in AFSC	explores the transformative potential of AI in the whole AFSC, focusing on its implications for agricultural productivity and consumer behavior	AFSC	literature review	the cluster are: adoption of AI technologies in agriculture, applications of AI in the intermediary management of the supply chain, consumer perceptions and acceptance of AI-powered food innovations, and the role of AI in precision nutrition and personalized health management	PRISMA	time scale	NA	NA
06	Halder et al. (2025)	AI-enabled secure social industrial IOT	analysis the AI-enabled secure social industrial IoT applications in AFSC, addressing key security concerns and efficiency bottlenecks	AFSC	literature review	AI-driven security solutions significantly enhance trust management, anomaly detection, and data privacy in social industrial IoT networks	taxonomy development	time scale, sample	NA	NA
07	Yohannis et al. (2025)	contamination in AFSC	synthesis of recent advances in technologies to mitigate contamination in AFSC	AFSC	literature review	integration technologies enable a transformative paradigm shift from reactive monitoring to predictive and preemptive of contamination	critical synthesis	data complexity,	cereals, groundnuts, oil seeds, and spices	tropical and subtropical regions

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08	Cordeiro and Ferreira (2025)	decentralized identity and digital twins for verifiable product identity	identify the impact of decentralized identifiers, digital twins, smart contracts, and blockchain to create verifiable digital representations of agricultural products and automate trust mechanisms	AFSC	literature review	blockchain technology enhances product identity, traceability, ability to create secure tamper proof digital records, maintains a verifiable history, increases transparency and reduces fraud	PRISMA	time scale	NA	NA
09	Sun, C. Zhao, and Xue (2025)	mode selection and logistics integration strategy	determine optimal sales mode and logistics strategy for Green Agro-Food Supply Chain (GAFSC) in e-commerce	GAFSC	mathematical modeling	in the reselling mode, when the logistics service fee is low, the retail price and the level of logistics service under the self-operated logistics strategy are higher than under the third-party logistics strategy	game-theoretic analysis	generalization	NA	China
10	X. Hu (2025)	AFSC network design	design GASCN and minimize total transportation costs	GASCN	mathematical modeling	ESSA outperforms the Genetic Algorithm and Particle Swarm Optimization in fitness and execution time, lower costs and faster convergence	Enhanced Spring Search Algorithm (ESSA), Genetic Algorithm (GA), Particle Swarm Optimization (PSO)	single-objective (cost)	NA	China
11	Remijnse et al. (2025)	management of food waste	develops a mixed-integer optimization model to support decision-makers in determining an optimal product portfolio and processing configuration	GAFC	mathematical modeling	side-stream valorization is generally well aligned with profit maximization, while it is not always beneficial from an environmental impact perspective	bi-objective mixed-integer programming	generalization	carrot	Netherlands
12	Meng, Fan, and X. Xu (2025)	information collaboration in AFSC	constructs a quadripartite game model involving the government, an information service platform, farmers, and agri-food enterprises. Analyzes stakeholders' strategic interactions and evolutionary pathways of stakeholders while exploring the regulatory effects of key parameters in reward and penalty mechanisms on system convergence	AFSC	hybrid approach	standalone fiscal subsidies can rapidly initiate the coordination mechanism, but their incentive effects are difficult to sustain over time	game theory, dynamics simulation	lack of accounting for external uncertainties	NA	China

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13	F. R. Taghikhah et al. (2025)	digital capabilities and their impacts on AFSC resilience	identify the nature and impacts of firms' digital capabilities on engineering versus socio-ecological resilience	AFSC	mixed methods	AFSC members' digital capabilities play distinct but interconnected roles in enhancing resilience, with their impact varying across persistence (engineering), adaptation, and transformation (socio-ecological) over different time frames	qualitative and analytic methods (comprehensive taxonomy)	non-generalization of the results	grains, red meat, dairy, horticulture, seafood, and wine	Australia
14	Muyulema-Allaica et al. (2025)	Key resilience and sustainability drivers and controllers in AFSC	identify and prioritize the key drivers and enablers of resilience and sustainability in AFSCs	AFSC	mixed approach	the main drivers are "Evolution of Agricultural Systems" and "Water Footprint." The main enablers are "Redesign and Coordination of Operations" and "Collaboration Across Supply Chain Links"	SLR, AHP, DEMATEL	lack empirical validation, limited generalization	NA	NA
15	Elkoraichi, Elfezazi, and Belhadi (2025)	blockchain in AFSC	identify barriers to blockchain adoption in the AFSC and effects for policy guidance	IAFSC	empirical and case study	12 barriers identified	DEMATEL analysis, expert opinion	small expert sample	NA	Africa
16	Farina et al. (2025)	technologies for enhancing transport systems	develop a model that enables the seamless integration of data from sensors, IoT devices, data management platforms, and distributed ledger technologies (DLT) within a newly designed data space architecture	UAFC, PSC	empirical and case study	platform improve data integrity, provenance, transparency, actionable ML insights from sensor data	design platform	generalization	cattle and livestock	Italy
17	Smyth et al. (2025)	AI in AFSC	examine AI-enabled information processing to build resilience in AFSCs via organizational mindfulness and flexibility	PAFSC	empirical and case study	organizational mindfulness and flexibility play key roles in building resilient supply chains, boosting long-term performance, and reducing food waste	generalization	NA	Europe	
18	Silva et al. (2025)	Sustainable agricultural practices transfer	examine how SUSAPs are implemented and transferred across multi-tier SC using a socio-material, practice-based lens	UAFC, FPAFC	empirical and case study	animal welfare fully established as a practice; waste management, working conditions, biosecurity remain proto-practices with partial/weak links, first-tier supplier as boundary spanner	interviews , qualitative methods	generalization	poultry products	Brazil
19	Mardenli et al. (2025)	Act on corporate due diligence obligations	assess perceptions and effectiveness of SC using expert perspectives	AFSC	empirical and case studies	large established companies are perceived to be less affected due to their existing transparency and compliance infrastructure	semi-structured interviews	sample	NA	Germany

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20	Jayalath, Ratnayake, et al. (2025)	Digitalization of AFSC	develop a digitalized, regulated architecture to transform SCs in developing economies and improve efficiency, transparency, and food security. Evaluate stakeholder interactions	UAFC	empirical and case study	the proposed framework offers benefits to all stakeholders in all supply chains while improving transparency, optimizing resources, leading to improved production planning, minimizing post-harvest wastage, reducing price volatility, and increasing food security.	reasoning approach	single country focus	seeds, and vegetables	Sri Lanka
21	Ranjbari et al. (2025)	Circular economy startups	develop a circular economy comprehensive typology that identifies their key characteristics and contributions to the CE	UAFC	empirical and case study	three distinct groups identified biotech-driven sustainable producers, digital CE enablers, and circular packaging pioneers	content analysis, hierarchical clustering	European focus	NA	Europe
22	Rackl and Menapace (2025a)	Geographical indication certification	examine the role of Geographical Indication (GI) certification in coordinating small- and medium-sized food suppliers and large-scale retailers in AFSC, where retailers seek to procure high quality goods	AAFSC	empirical and case study	The results suggest a potential role for GI certification in supplier-retailer coordination	survey, SLS, GMM, PSM	generalization	Food craft (bakery, butcher, confectioner, miller, brewer, cheese dairy, and winemaker)	Germany
23	Chud-Pantoja, Manyoma-Velasquez, and Soto-Paz (2025)	Circular economy in AFSC	selection of AFSCs with circular economy potential, through the implementation of quantitative and qualitative prioritization, proposing indicators related to the circular economy	UAFC	multi-criteria analysis	top prioritized products: avocado, grapes, guava, citrus	Shannon and Condorcet index	regional single region	44 products (fruits/vegetables)	Colombia
24	W.-H. Yang, Y.-C. Chen, and Y.-J. Yang (2025)	Severity of losses and wastes	evaluate the severity of losses and wastes	UAFC	Multicriteria analysis	the eight severe issues identified “Inadequate planning and control of overall production and marketing policies” under government policies, “Adverse climate conditions” and “Imbalance in production and marketing” under production site loss, “Inaccurate market demand forecasting” and “Poor inventory management at supermarkets” under supermarket waste, and “Improper storage management of ingredients leading to spoilage” and “Inability to accurately forecast demand due to menu diversity” under restaurant waste.	BWM, expert survey	qualitative weighting with expert subjectivity	NA	Taiwan

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25	Gupta et al. (2025)	Blockchain implementation challenges	identify and evaluate the main blockchain challenges affecting food SMEs	AFSC	Multi-criteria analysis	lack of management commitment, negative perception of BCT, and high implementation costs emerged as the primary obstacles to BCT adoption in AFSC	literature review , expert questionnaire , IMF-SWARA, TFBM approach	small expert sample	NA	India
26	Jayalath, Perera, et al. (2025)	traditional AFSC	evaluates the integration of circular economy practices to enhance sustainable production in order to reduce waste in traditional AFSC in developing economies	AFSC	multi criteria analysis	information flows are the most impactful flow in minimizing supply chain waste, followed by material and financial flows	interviews, AHP, and ISM/ANP	single-case	vegetables	Sri Lanka
27	Yildirim, Ayyildiz, and Aydin (2025)	sustainable strategies in AFSC	Assess sustainability AFSC	UAFC	multi-criteria analysis	low R&D investment is the major barrier while adopting digital technologies emerges is the most effective solution	SF-PIPRECIA , SF-CODAS , SF-TOPSIS/WISP , SWAM	single-country case	Tea	Turkey
28	Shahid et al. (2020)	Blockchain in AFSC	End-to-end blockchain-based AFSC solution with traceability, trading, delivery, reputation, and security	AFSC	mixed approach	Demonstrates traceability, accountability, credibility, and autonomous delivery with vetted gas costs and security analyzes	Smart contracts , IPFS storage, off-chain arbitrator	scalability	NA	NA
29	Ciccullo et al. (2021a)	role of food waste prevention technologies	investigates the range of the available technologies and the detailed objectives of such technologies for food loss and waste prevention	pre-harvest to post-harvest	mixed approach	adoption of different technological options can represent the engine to establish vertical collaborations between the adopter of the technology and another stage in the AFSC, in order to fight food waste and loss with a coordinated supply chain effort	interviews, qualitative taxonomy development	generalization	fruit and vegetable	Italy
30	Saurabh and Dey (2021)	Blockchain in AFSC	identify the potential drivers of blockchain technology adoption	AFSC	mixed approach	disintermediation, traceability, price, trust, compliance, and coordination and control can influence the supply chain actors' adoption-intention decision processes	conjoint analysis	Generalization	grape wine	India
31	Mishra, Rajesh Kumar Singh, and Subramanian (2022)	disruptions in AFSC	study disruptions in AFSC during COVID-19 and develop capabilities for resilience and operational excellence using SAP-LAP and dynamic capabilities	DAFSC	mixed approach	propositions linking disruptions, capabilities, and resilience, framework showing proactive/reactive strategies, actionable actions for demand/supply/logistics	SAP-LAP framework, interviews, multiple data sources, triangulation	single case, generalization	perishable products	India
32	Taylor (2006)	development of lean AFSC	present an initial model of an integrated supply chain based on the application of lean principles	PAFSC	empirical approach	value chain analysis methods combined with lean principles can provide a powerful framework for the analysis and improvement of supply chain activity	value chain analysis (VCA)	two case studies within a single sector	pork product	UK

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33	Leat and Revoredo-Giha (2013)	risk and resilience in AFSC	identify the key risks and challenges involved in developing a resilient agri-food supply system, particularly about primary product supply, and to show how risk management and collaboration among stakeholders can increase chain resilience	PAFSC	Empirical approach	horizontal and vertical collaboration reduced vulnerability, improving market security for producers and supply reliability for retailers	qualitative case study	single specific supply chain analysis	pork products and Pigs	Scotland
34	Rana, Tricase, and De Cesare (2021)	Blockchain technology	realize a SLR on blockchain technology applications for sustainable AFSC, analyzing benefits and challenge	AFSC	literature review	BCT improves sustainability, traceability, and food safety, but faces challenges like scalability, privacy leakage, high costs, and connectivity	SLR	search criteria and keywords excluded	NA	NA
35	Bakhshi Sasi, Sarker, and Essam (2024)	mitigation in a AFSC	propose a novel heuristic algorithm that revises the ideal plan in the event of a disruption	GFSC	mathematical approach	proposed heuristic is highly accurate (gap vs. exact methods) and significantly faster	MILP-H	study is deterministic and limited to single period	rice and wheat	Global
36	Karki (2023)	Closed-Loop supply chain network design optimization	propose a sustainable closed-loop fruit supply chain using the Red Deer Algorithm	closed-loop AFSC	mathematical approach	Red Deer Algorithm (RDA) outperformed other methods (GA, SA, KA, ICA) in solution quality and efficiency	RDA, GA, SA, KA, ICA	higher computational time (CPU) for large-scale problems	NA	
37	Wen and Mo (2025)	impact of environmental regulations and digital empowerment	Understand environmental regulation and digital empowerment on green AFSC under uncertainty using a tripartite stochastic evolutionary game	GAFSC	mixed approach	Existence of multiple equilibria; digital tech and regulation synergistically drive green transformation	Stochastic evolutionary game theory , Gaussian white noise	generalization	chicken	China
38	Kafi et al. (2025)	AFSC Research Directions	Identification of future research directions	AFSC	literature review	7 main research clusters identified: AFSC system, AFSC management, AFSC industry, AFSC risk factors, AFSC information, AFSC advancement, AFSC risk management	bibliometric analysis	recent publications not reviewed (until 2021)	NA	NA
39	Rohit Sharma, Sundarakani, and Manikas (2025)	role of I4Ts in enhancing the resilience of AFSCs	analyze resilience enablers and assess industry 4.0 technologies (I4Ts)	AFSC	literature review, integrated methodology, interviews	the most influential resilient enablers: real-time information sharing, enhanced product traceability, improved risk management, and planning and network design; I4Ts enhance the resilient enablers: big data analytics, Internet of things, and cloud computing	GINA: identify the most influential resilience enablers; FLQOWA: assess and prioritize I4Ts to enhance resilient enablers	no verification of causality of factors, absence of data-cleaning, non-generalization of the results	NA	United Arab Emirates

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40	M. Khan et al. (2025)	role of intermediaries in the sustainability of AFSC	assemble all the stakeholders of an AFSC into a coordinated platform	MTAFSC	single case study methodology	this integration has contributed to increase actors' trust, collaboration at different levels	qualitative case study method is based on semi-structured interviews	lack of evaluation of the economic impact of the sustainable practices on costs in MTAFSC	rice	Pakistan
41	Perçin, Bayraktar, and V. Kumar (2025)	integration of industry 4.0, sustainability, and circular economy (I4.0-S-CE) into AFSC	identify key I4.0-S-CE drivers and the prioritized order of importance of agricultural sustainable development goals	AFSC	literature review, expert opinions, case study	the most important sub-drivers: resource efficiency, economic and social sustainability, and the achievement of standards and SDGs . The effective use of resources is considered the most important key to meeting the agricultural needs of countries. Sustainable consumption and production are the most crucial targets. Then, hunger, clean energy, and life on land will guide countries towards the goals of zero hunger, renewable and clean energy, and ecosystem protection. Furthermore, other agricultural SDGs are essential, such as eradication of poverty, improved quality of life, protection of water resources and mitigation of climate change	hybrid Fuzzy MCDM Method (FBWM, FTOPSIS, FARAS, FSAW)	not taken into account the peculiarity of developing countries (insufficiency of technological infrastructure, lack of cooperation between stakeholders, and limited awareness of sustainability, circularity and regulatory)	Oils, dry food, bakery products,canned products,drinks	developing countries (Türkiye)
42	Veloso et al. (2025)	assessing circular economy in AFSC	identify strategies that promote EC to various stages of the AFSC, identify a classification of sustainability KPIs	AFSC	case research methodology (analysis of sustainability reports)	31 indicators identified for measuring, supporting, or controlling CE strategies+6 new standards proposed: avoided food waste; food waste identification and management; reverse logistics; circular packaging design; and regenerative farming	expert focus group, and practitioners' interviews	research not exhaustive, limited sample	NA	Portugal
43	Obayi et al. (2025)	Intersection of industry 4.0 technologies and human–machine value	explore how system and business-level interoperability at the' motivations	Agri-food digital ecosystem	qualitative research approach	system-level is driven by defensive strategies for compliance and standardization, while the interoperability at the system level relies on offensive strategies for flexibility and innovation	case study, qualitative analysis (interviews, field observations and focus group discussions)	generalization (focused on one digital ecosystem in India)	NA	India

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44	G. Zhao, Ye, Zubairu, et al. (2025)	Adopting I4.0 technologies in AFSCs	identify barriers to the adoption of I4.0 in AFSCs.	AFSC	fuzzy analytic hierarchy process	the key challenges to applying I4.0 are the increased cost of terminal logistics, acquisition of intelligent agricultural equipment subsidies based on Guanxi, low compatibility of I4.0 technologies with existing agricultural equipment, and problems with the government subsidy model	literature review , consultation of experts and Multi-criteria tool	Generalization, limited scope of literature and few expert input.	NA	china
45	Lipińska (2024)	Protection of the position of an agricultural producer in AFSC	determining how the legal instruments provide protection for agricultural producers against unfair trade practices to strengthen the position and increase the competitiveness of the agricultural producer	PAPSC	qualitative and legal methodology	strengthening the position of farmers in the food supply chain is one of the key objectives of the Common Agricultural Policy (CAP). It can be achieved through the development of cooperation among the actors of the chain (agricultural producers)	legal analysis, comparative study, interpretation	study focused on the analysis of legal texts, without direct treatment of the concrete effects on producers	NA	NA
46	Tebaldi, Gubiotti, and Vignali (2024)	quantitatively assessing the resilience level	propose an analytic quantitative model to quantify the resilience level of an AF system	AFSC	analytic model, case study	23 qualitative and quantitative indicators identified for a resilient AFSC	literature review, descriptive statistics method	Employing in-depth interviews, study geographically limited	coffee	Ethiopia
47	Habib et al. (2024)	role of critical factors in the durability of AFSC	Identify drivers, enablers, and barriers to adopting sustainable practices	AFSC	literature review(identifying factors) and empirical survey(ranking factors)	The key critical factors : economic and productivity improvement, cost effectiveness and improvement in overall performance, and difficulty in mindset and cultural changes	descriptive statistics method (analyze data+ranking factors)	Employing in-depth interviews, study geographically limited	coffee	Ethiopia
48	El Hajji et al. (2024)	Optimization AFSC using platform based on blockchain	propose a web platform "Hyperledger Fabric" for dynamic real time visibility of a package's movement	AFSC	platform-based design and evaluation	real time visibility of a package's movement and security of the AFSC	cryptographic algorithms (MSP and CA), evaluation metrics, benchmarking results	the lack of predictive insights which can be resolved by on integrating IoT and advanced analytics	NA	NA
49	G. Zhao, Ye, Dennehy, et al. (2024)	Obstacles to adopting Industry 4.0 for a sustainable AFSC	identify barriers and prioritize them	AFSC	SLR, structured interview+ group-based fuzzy analytic hierarchy process (GFAHP)	The three main barriers are I4.0 technologies that increase the cost of terminal logistics, the acquisition of subsidies for smart agricultural equipment and the incompatibility of technologies with existing agricultural equipment	literature review, survey	generalization, limit to papers published in high-quality journals, three experts	NA	Chine

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50	Ricciolini et al. (2024)	Sustainability of European AFSC	measure the sustainability level (by countries and sectors: agriculture, industry, distribution and consumption)	AFSC	multi-criteria analysis	Italy ranked highest in AFSC sustainability, followed by Sweden, Austria, Spain, France, Germany, Czech Republic, Portugal, and Slovakia. The most significant sustainability issues were observed in the consumer sector	Multiple Reference Point Partially Compensatory Indicator MRP-PCI	generalization	NA	European Union
51	Rashi Sharma et al. (2024)	selection of performance indicators for sustainable supplier evaluation	propose a group consensus decision making model based on an iterative algorithm to generate weights, select suppliers who can embed factories 's sustainability requirements into their SC operations	AFSC	multi-criteria decision-making	the top ranking indicator for sustainable supplier evaluation research is 'Food Safety' then 'Payment to Sub-suppliers'	Delphi method, GAHP, Kendall's concordance coefficient, case study	ranking of performance indicators may change according to the decision-making environment	flour milling	India
52	Belhadi et al. (2024)	Management of AFSC uncertainties following geopolitical disruptions	study the role of digital dynamic capabilities in enhancing SCRes in AFSCs impacted by geopolitical disruptions	AFSC	multi-case study approach	strong relationship between the dynamic capabilities of digital supply chain technologies and SCRes at both organizational and inter-organizational levels, proactive and reactive SCRes are built on digital capabilities	qualitative, exploratory research, multi-case study design, interpretive analysis, exploratory interviews, semi-structured interviews, cross-case analysis	generalization (14 firms)	NA	Africa
53	Rackl and Menapace (2025b)	Coordination in AFSC through geographical certification	identify the impact of geographical indication certification for vertical coordination	AAFSC	Interviews, empirical study	GI certification enhances the coordination of procurement contracts by improving the ability of the retailer to identify efficient suppliers and encouraging the provision of high-quality goods	econometric model	non-generic results depending on the product type and country	food craft	Germany
54	Chandrasiri et al. (2024)	Adoption of bi-modal transport for reducing food miles	reduce transport distances (food miles and and empty haul distance), transport cost and emission footprints by modifying the volume and direction of the product flow	DAFSC	analysis model, case study	bi-modal transport allow the reduction of food miles by 32%, transportation costs by 36%, contributions to the global warming potential by 35%, and empty truck hauls by 38%	mixed-integer linear programming model	single-objective linear programming, generalized constraint (different categories of vegetables, storage condition, shelf-life time of vegetables,etc)	vegetables	Sri Lanka

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55	Gámez-Albán, Guisson, and De Meyer (2024)	vehicle routing problem (problem of the pickup round: first mile): Heterogeneous Fleet Routing Problem (HFRP)	minimize total logistics costs of visiting fields during a pickup round using multiple vehicles.	PHL	analytic model+case study	organizing the first mile with a cooled vehicle for pick-up rounds reduces logistics costs by up to 40% % compared to other practices	mixed-integer linear model	Single-objective linear programming (e.g. lack of study of maximization of product freshness at the time of sale)	peaches and nectarines	Greece
56	Aghajani et al. (2024)	optimization AFSC	maximize the net value of the whole supply chain, maximize the job opportunity, minimize the greenhouse gas emission (minimize overall cost at various stages, maximize the revenue and employment generated and reduce the environmental impacts, maximize job opportunities in the chain by maximizing the man hours needed while reducing the greenhouse gas emission, propose a pricing model to estimate the price in various customer zones)	AFSC	literature review, mathematical modeling, sensitive analysis, robust possibilistic programming	temperature exert the highest impact, whereas land use shows the lowest	multi-objective integrated mathematical model	model limited to a single case study and expert-based assumptions.	stevia	Iran
57	Alif et al. (2025)	Circular agri-food supply chains, zero-waste transition, sustainability, optimization and stakeholder collaboration.	Identify research trends and optimization advances	AFSC	systematic review;	Economic/environmental dimensions dominate; social sustainability, AI/ML integration, interdisciplinary modelling and real-world validation remain insufficient.t	PRISMA	limited reviews	NA	multi-country

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58	Arimany-Serrat, Montanyà, and Amat (2024)	causal relationship between resilience and sustainability	identify the sustainability variables that help the resilience of companies	RSC	qualitative comparative analysis methodology, case study	the resilience of supply chains is strongly linked to strategies to improve environmental, social and governance sustainability (pollutant emissions, safety and health of workers, diversity in management bodies and employees, reduction in energy consumption, selection of suppliers with social criteria and fight against corruption)	investigation	sample of eight distribution companies	beverages, frozen foods, dairy, dry foods and grocery, meat and charcuterie, pet food, home care, personal care	Spain
59	Sakthivel et al. (2024)	Blockchain enhances trust and traceability	propose a blockchain architecture	CAFSC	experimental methodology	blockchain enhance managing a massive number of data and transactions, avoiding fraud, securing records, strengthening trust	architecture, deployment, evaluation	lack of traceability	NA	NA
60	Charlebois et al. (2024)	digital traceability adoption and its impact in AFSC	provide an overview of countries' efforts in digital food traceability and propose actionable recommendations and insights for effective deployment of digital traceability systems	AFSC	comparative analysis, multidimensional analytical framework	OECD countries increasingly adopt digital traceability especially blockchain and QR codes enhancing food safety and transparency, while highlighting disparities, regulatory gaps, and the need for standardized framework	case studies and empirical study(quantitative analysis, regression analysis)	incomplete data, generalization	NA	38 Organization for Economic Cooperation and Development (OECD) member countries
61	Sayma et al. (2024)	reason of price hike in AFSC utilizing private Ethereum blockchain network	propose a system that enables tracing the origin of corruption in the supply chain and identifying the source of price increases	AFSC	conceptual framework, system design, implementation	the proposed system ensures a high level of transparency, as every transaction recorded on the blockchain is visible to all network actors, thereby promoting openness and reducing information asymmetry. it provides strong traceability, allowing each transaction to be tracked back to its specific Ethereum address. This feature facilitates Data Protection , verification, Fraud Prevention,Security ,Transaction speed and Cost Efficiency across the entire supply chain.	private Ethereum blockchain, Proof of Authority (PoA) approach, InterPlanetary File System (IPFS), dApp (decentralized Application)	simple and no complex AFSC	vegetables (potato and eggplant)	Bangladesh

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62	G. Wang, S. Li, Yi, et al. (2024)	Impact of digital technology in the sustainability of CBAFSC	identify the research hotspots and the potential research directions	CBAFSC	bibliometric analysis	digital technology enables sustainability and food safety in CBAFSC	bibliometric analysis with visual knowledge map	sample literature, interpretation with "Citespace ", without comparing and validating the results with other literature analysis software (eg. VOSviewer , Histcite)	NA	NA
63	El Bhilat, El Jaouhari, and Hamidi (2024)	Impact of AI on AFSC performance	examine the effect of AI based technologies on reducing wastes and minimizing AFSC costs, hence increasing organization profitability	AFSC	conceptual model, Harman statistical test, Partial Least Square Structural Equation Modeling (PLS-SEM), confirmatory factor analysis (CFA), quantitative approach, common method bias (CMB), hypothesis test (survey), dynamic capability view (DCV) and organizational information-processing theories	distribution network efficiency ,AFSC efficiency, and organization performance are directly and positively affected by AI integration. Efficient distribution logistics in the agri-food sector reduces product waste, ensures timely delivery of perishable goods, and improves response to market changes.	Partial Least Square-Structural Equation Modeling (PLS-SEM) approach	study based on survey study, generalizability	NA	Morocco
64	Camel et al. (2024)	integration of BCT with Green Product Platforming (GPP) in the agri-food industry	analyze the impact of BCT-enabled Green Product Platforming on reducing the carbon footprint and enhancing economic performance	AFSC	quantitative surveys, qualitative interviews, data analysis	the importance of aligning GPP initiatives with stakeholder expectations and leveraging BCT to navigate and capitalize on stakeholder pressures for improved carbon and economic performance	partial least squares structural -equation modeling (PLS-SEM).	sample, generalization	NA	NA

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65	Hamidoğlu (2024)	government-backed agri-food supply chain model (GBASM) between farmers and agricultural enterprises	assess the impact of the GBASM on the decision-making processes of farmers and agricultural enterprises (supplier, manufacturers/factories, distributors, supermarkets/wholesaler/ retailer)	GBAFSC	Game-theoretical approach + case study	the proposed model allows to achieve affordable market prices among all relevant actors and establish efficient pricing mechanisms within their market environments	Nash equilibrium +numerical experiments, grey wolf optimizer algorithm	Increased complexity (increasing the number of actors makes the Nash equilibrium more difficult to establish)	tomato	Argentina
66	P. Liu, B. Zhao, and Q. Liu (2024)	Impact of government tax subsidies on Blockchain technology investment strategies in AFSC	analyzes tax subsidies for blockchain adoption and updates the demand function based on consumer trust	GAFSC	Game-theoretic modeling	If blockchain investment costs are within a certain range, tax subsidies yield higher stakeholder benefits. However, full subsidies do not always raise all actors' incomes, as this depends on subsidy rates and the blockchain technology investment cost	Mathematical modeling, reverse analysis (master-slave game), Numerical simulation	not distinction between online and offline channel	cherries	China
67	Guidani, Ronzoni, and Accorsi (2024)	deployment of a digital twin	Virtualize AFSC processes to estimate economic, logistical, environmental, safety, and nutritional indicators for every food order.	AFSC	deployment	the digital twin provides process visibility and helps practitioners make operational and tactical decisions through real-time monitoring and multidimensional post-performance analysis. It also enables consumers, through their informed choices, to exert strategic pressure on the food industry to ensure sustainability.	Relational modeling (SQL) +simulation +OO paradigm +Monte Carlo method	limited simulation analysis (performed on four virtual scenarios)	nine different horticultural products	Italia
68	D.-C. Toader, Rădulescu, and C. Toader (2024)	Blockchain technology in AFSC	analyze the factors exerting a positive effect on blockchain adoption within AFSC	AFSC	empirical study	the significant influence of performance expectancy , effort expectancy , and perceived trust on usage intention, while also revealing the positive impact of Organizational Blockchain Readiness (OBR) on expected Usage Behavior (UB)	Unified Theory of Acceptance and Use of Technology(UTAUT)) , Partial Least Squares Structural Equation Modeling (PLS-SEM), investigations (questionnaire), Common Method Bias (CMB)	Sample, generalizing, non-exhaustive study factors	livestock and fresh products	European countries (Czech Republic, France, Germany, Italy, Poland, Romania, Spain, and Switzerland)
69	Marín, Garnatje, and Vallès (2024)	impact of AFSCs in biodiversity of in peri-urban agricultural Areas	analyze the role of CFSc in conserving and promoting local agro-biodiversity	CFSC	case study approach	distribution plays a crucial role in local food systems, not only involves the trade of food but also influences consumption patterns, societal values, and the overall landscape	semi-structured interviews, inventory	local scope, limited sample	fruits and vegetables	Catalonia (Spain)

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70	Huilan Chen et al. (2024)	influence of critical factors on the success of knowledge mobilization	identify the critical success factors and the correlation between them	IAFSC	quantitative analytical methodology	8 of the identified criteria are strongly correlated with the success of knowledge mobilization (collaboration, structure of the supply network, power, technology, trust, commitment, training/education, time, cost, culture, and continuous improvement), 3 of them disappeared (continuous improvement, time and cost)	questionnaire surveys, multiple regression analysis	rank the factors using more scientific methods, generalization, model based on literature	NA	NA
71	Cossu et al. (2024)	Eco-labelling and sustainability AFSC	identify the main impact categories of the environmental footprint of the product life cycle.	AFSC	Empirical Methodology(Case studies+ analysisis	The score reflects an intermediate environmental performance +the role of eco-labeling in encouraging more sustainable practices in the sheep dairy phase +the crucial role of sheep farming systems for improving the environmental footprint .	PEF method+ experimental MGI (Made Green in Italy) application+ ACV	sample, generalization	hard sheep milk cheese	Italy
72	Stevens and Teal (2024)	impact of diversification firms in resilience AFSC	develop revenue-based measures of firms' vertical (across supply-chain segment) and horizontal (within supply-chain segment) diversification. Compare post pandemic outcomes of more diversified and less diversified firms	AFSC	empirical approach	vertical diversification reduces firms' resilience, whereas horizontal diversification increases firms' resilience, horizontal diversification of small and medium sized firms increases resilience in the AFSC	inverse probability weighting (IPW), CBGPS and NPCBGPS techniques, survey	sample	NA	U.S (California, Florida, and the two-state region of Minnesota–Wisco
73	Ikasari, Suef, and Vanany (2025)	risk management and sustainability in the AFSC	identify the development of research topics and the future research opportunities	AFSC	bibliometric analysis	increasing trend in publications on risk management, sustainability, and AFSC over the past six years, future research shoud focus to the impact of emerging technologies in resilience and sustainability supply chain	bibliometric analysis	single Scopus, based database	NA	NA
74	Chu and Pham (2024)	blockchain-enabled vertical coordination in AFSC	develop a blockchain-based framework for improving vertical coordination in AFSC	IAFSC	literature review, framework, case study	blockchain technology provides benefits for each supply chain actor in terms of operational effectiveness, cost and time savings, and improved human resource	conceptual framework analysis	No real-world implementation or empirical validation of the proposed framework	cashew nut	Vietnam
75	Annane, Alti, and Lakehal (2024)	blockchain for Secure and Traceable AFSC	propose lightweight blockchain-based signature mechanism for rapid detection of food fraud	AFSC	Semantic-based Approach	$\tilde{7}$.5 speed improvement over iterative search algorithms and maintaining high transaction traceability with significantly reducing storage costs	systematic analysis, semantic approach	no privacy-preserving technologies, no advanced detection of unknown attacks	NA	Algeria

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76	Kim, Ivanov, and Yoon (2025)	improving product safety	examine direct investment in supplier development and the adoption of blockchain technology for enhanced traceability and transparency, the impact of an increased inspection frequency on food safety, verification the current government interventions (increasing inspection frequency and raising quality standards)	IAFSC	qualitative approach and game theory analysis	both direct investment and blockchain technology adoption can independently promote agrifood product safety, blockchain technology adoption is effective when the retailer secures a low unit margin (i.e., a high wholesale price)	intensive interviews, Nash equilibrium	lack of demand uncertainty of the retailer and considering competition with other retailers	NA	Korea
77	Barbieri et al. (2025)	the assess the environmental performance of the logistics distribution processes	identify key elements for evaluating the environmental impact of distribution processes	AFSC	empirical-analytical simulation approach	intermodal transport as more environmentally sustainable than full road transport. Potential CO2 emission savings of up to 50% through improved vehicle efficiency and optimized vehicle utilization. Additional savings of up to 30% through increased use of intermodal transportation.	semi-structured interviews, algorithmic model, business case	limited scope	NA	EU
78	Otter and Robinson (2024)	digital transparency in AFSC networks	identify the primary and secondary stakeholders in the AFSCNs of the digital era	AFSC	empirical study	11 secondary net chain stakeholders must be integrated in information AFSC network	qualitative semi-structured interviews with technology providers, structured analysis	lack of assessment of the importance and interests of stakeholders	NA	EU
79	Yin et al. (2025)	relationships between AI, AFSC management practices, resilience, and performance	study the impact of AI on AFSC resilience, and performance	PAFSC	analytical approach, empirical study, statistical analysis	the top organizational factors play a crucial role in ASC resilience through management support (TMS), supplier integration (SI), internal management (IM), customer integration (CI), and supply chain collaboration (SCC). The AI moderates these relationships, AI improves internal management practices due to advanced analytical capabilities for effective decision-making (prediction, and improvement of communication, efficiency and coordination)	partial least squares structural equation modeling (PLS-SEM), survey , Harman factor, analysis of reliability metrics	non generalizable	NA	China

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80	Moreno-Miranda and Dries (2026)	correlation between coordination mechanisms and sustainability performance	evaluate the role of coordination and transaction costs in the sustainability performance SC, examines the interrelations between transactional characteristics, relational and formal coordination, and sustainability performance in AFSC	AFSC	exploratory and quantitative study, statistical analysis	relational governance improve economic and social sustainability, enhance indirectly ecological performance via resource efficiency+, formal coordination and effective transaction cost management enhance collaboration and supply chain viability	survey, structural equation modeling, confirmatory factor analysis, exploratory and confirmatory factor analyses, structural equation modeling, validity and reliability testing, structural equation model (SEM), Promax Rotation	limited number of variables to measure sustainability performance	blackberry	Ecuador
81	Paredes-Rodríguez, Orejuela-Cabrera, and Osorio-Gómez (2024)	sustainability and resilience in AFSC	identify the relation between resilience and sustainability	AFSC	literature review	resilience and sustainability as two cross cutting and highly interrelated concepts in supply chain management	systematic reviews and Meta-Analyses (PRISMA protocol)	content analysis sample of papers, lack of a robust relationship	NA	NA
82	Loy, Britton, and Malone (2024)	environmental impact reporting and Software solutions in AFSC	determine the importance of developing and implementing advanced supply chain management software to address the need for improved sustainability practices in AFSC, identify the SCM software is being used for enhance sustainability, determine the impact of software skills on sustainability reporting and governance	AFSC	qualitative research approach	the adoption of software allows for a significant reduction in data collection costs, but human intervention remains essential to manage technological failures	semi-structured, qualitative interviews	limited sample size	NA	United States

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83	Israel and Siwandeti (2024)	farm economic viability	identify the interaction between agricultural supply chain capabilities (AGSC), dynamic pricing (DY PG) and subsidy schemes (SBSM) and their collective influence on farm economic viability	LAFSC	empirical approach	smallholder farmer face various challenges that impact their agricultural supply chain capabilities, generating significant increases in food market prices (eg. increased costs related to transportation, storage, and losses)	survey, structural equation modelling (SEM) , structural equation modelling (SEM), descriptive statistics and correlation analysis, pearson correlation, moderated mediation analysis	generalization	staple food products (millet, wheat, barley, maize, rice, and sorghum))	Tanzania (Dodoma, Morogoro, Mbeya, Songwe, and Tabora)
84	Hasan et al. (2024)	IOT on enhancing transparency and efficiency of AFSC	define the influence of IoT on Physical and Information Flow(PHF) and the impact of an IoT-enhanced PHF on chain transparency, this helps promote better decision-making, efficiency and consumer confidence	AFSC	experimental design approach, hypothesis Testing and Path Analysis	significant and positive impact of IoT technologies on the PHF and on the transparency of AFSC	sample (probability simple random sampling), exploratory factor analysis, confirmatory factor analysis (CFA), structural equation modeling (SEM)	generalization, sample size	NA	Bangladesh
85	González-Mendes et al. (2024)	key issues and trends of blockchain in AFSC	identify future research and gaps in this topic	AFSC	bibliometric analysis	the research hotspots are related to the integration of blockchain technology in the AFSC for traceability, coordination, transparency and Enhancement of food safety	co-occurrence analysis, descriptive analysis, bibliometric analysis	limited database, time scale	NA	NA
86	Joshi, M. Sharma, et al. (2025)	role of Industry 4.0 in enhancement the sustainability of AFSC during the health crisis	assess the impact of digital technology 4.0 on sustainability of AFSCs	AFSC	quantitative research	industry 4.0 technologies have major impact on the viability of sustainability of AFSC. Organizational flexibility determines how Industry 4.0 impacts AFSC sustainability	IPT theory, structural equation modelling ,PLS-SEM, statistical analysis (measurement model, structural model, hypothesis testing)	limited sample (including companies in transition)	food grains	India

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87	Gholian-Jouybari, Hajiaghaei-Keshteli, Smith, et al. (2024)	design a sustainable closed-loop AFSC network	identify the optimal number and location of facilities, the flows (i.e., the number of goods) to be transported based on facilities' capacity, and the inventory level of products, minimizes the total cost, environmental pollution, and maximizes job opportunities	CLAFSC	depth metaheuristic	optimizers can solve all sizes of problems	non-dominated sorting genetic algorithm-II (NSGA-II), interval plots and the Friedman statistical test, six metaheuristics and three hybrid metaheuristics are utilized, Taguchi method, sensitivity analysis	deterministic data, static model	coconut industries	Mexico
88	G. Zhao, Vazquez-Noguerol, et al. (2024)	resilience strategies	identify resilience capability factors used at AFSC levels in the face of the crisis	AFSC	empirical approach	frequently discussed resilience capabilities, such as collaboration, redundancy, flexibility, leadership, and innovation, were implemented across the preparedness, response and recovery, and adaptation phases; however, successful AFSC recovery also depends on each country's cultural value orientations	middle-range theory, thematic and comparative analyses, middle-range theory (MRT)	limited geographic scope	NA	China and Spain
89	Baležentis et al. (2024)	effects of the disruptive events on the viability of AFSC	assess of AFSC viability during the disruptive external effects (effects of COVID-19 and the Ukraine war)	APC	probabilistic model	most of the negative factors arise from increased energy consumption in the agri-food sector. Positive effects are linked to the increase in production, which should associated with the ability of Lithuanian agri-food producers to maintain production activities uninterrupted	complex and sophisticated expert evaluation processing technique, power ordered weighted average (POWA) operator, Monte Carlo simulation	limited scope of actors	cereals, horticulture, fruit and berries, dairy, poultry, beef cattle, and pigs	Lithuania
90	Trevisan and Formentini (2024)	digital technologies for prevention and reduction of food loss and waste	analysis the state-of-the-art of adoption of each digital technologies for preventing and reducing FLW	AFSC	systematic literature review and research agenda	digital technologies are more frequently adopted in the earlier stages of the supply chain, IoT and digital platforms are the most frequently cited technologies, IoT is diffused throughout the entire supply chain	SLR	generalization	NA	NA

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91	Esteso et al. (2023)	improving the robustness in AFSC	design a a system dynamics (SD) model to analyze, measure and improve the robustness of an configuration and its planting planning to disruptions in demand, supply, transport between SC nodes, and the operability of SC nodes	PASC	mathematical formulation and simulation of scenarios	in the event of a disruption (logistics bottlenecks are the most critical), demand disruptions have immediate impacts on satisfaction and revenue, adaptation strategies (opening alternative nodes, adjusting workforce, reallocating logistics flows	known behavior replication test and the extreme conditions test, case study	generalization	vegetables	Spain
92	G. Zhao, Olan, et al. (2024)	links between risks and resilience capabilities	identify key risks and resilience capabilities in AFSCs, the relationships, correlations and causalities between them, and research gaps and future research directions in the field	AFSC	systematic literature review (SLR), comprehensive analysis	identify eight types of AFSC risk and seven types of AFSC resilience capability	content analysis	lack of an empirical study	multi-product	multi-country
93	G. Zhao, S. Liu, Y. Wang, et al. (2024)	modelling enablers	identify the resilience capability factors used to build AFSCRes, identify capability factors interrelated, identify resilience capability factors key to building AFSCRes	CBAFSC	Cross-impact matrix multiplication applied to classification (MICMAC) analysis, cross-country comparative analysis	contractual restraints regulating farmers' opportunistic behavior and regular interactions are key factors for building AFSCRes in France and Argentina, respectively	interviews, thematic analysis, total interpretive structural modelling (TISM)	validity of the empirical findings	organic Vegetables, tomatoes, cereals, seeds, agricultural inputs	Argentina and France
94	Kersten et al. (2024)	traceability in AFSC	investigate the traceability and its technologies	CBAFSC	bibliometric and content analysis(systematic literature review SLR)	traceability is essential at all levels of the supply chain, contributing to the management of food waste and sustainable practices. Various technologies are presented and a framework was developed to illustrate the application of these technologies at different stages of the supply chain and their relationships with the dimensions of ReSOLVE	systematic and structured process, PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)	uses only the ReSOLVE framework to analyze technologies	NA	NA
95	**KÃűrmendiovÃą85**	Vertical linkages in AFSC	identify the factors that influence the choice of distribution channel of family farms	FFFSC	statistical analysis	the choice of the distribution channel depends on the size of the company and the type of production. Businesses with less than 10 employees prefer direct distribution . Direct distribution channels are used primarily in the supply chain of animal products contrary to the indirect distribution chain specified mainly in the supply chain of plant products	questionnaire survey, statistical analysis, chi-square test	sample	NA	Slovakia

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96	G. Wang, Hou, and Shin (2023)	cross-border E-Commerce platforms in AFSC	identify and classify the key facilitating factors influencing the sustainable development of e-commerce platform	CBAFSC	systemic approach	the five most fundamental factors are simplification of customs clearance procedures, digital empowerment, consumer demand forecasting, knowledge innovation linkage, and policy environment optimization	Grounded Theory and DEMATELISM-MICMAC	non-taking of heterogeneity of platforms	NA	china
97	**KrstiÄĞ87**	e-traceability drivers	identify factors that motivate companies to implement electronic systems for tracking (e-traceability) and monitoring products	AFSC	multi-criteria decision-making (MCDM)	the most important drivers are supply chain efficiency, technology development and sustainability	"Factor Relationship" (FARE) and "Axial Distance based Aggregated Measurement" (ADAM) methods	sample of 27 experts	NA	NA
98	Gholian-Jouybari, Hajiaghaei-Keshteli, Bavar, et al. (2023)	design of AFSC network	develop a mathematic model to manage the total net profit while monitoring CO2 emissions and the satisfaction of customers within the network	CLAFSC	math	optimizers can solve all sizes of problems. However, MOGWO is better suited for addressing small-size problems, whereas MOHHSA is highly effective for tackling problems of medium and large sizes	mixed-integer linear programming model, conventional and hybrid meta-heuristics (four multi-objective optimizers and three hybrid algorithms are)	lack of reliable database, environment uncertain not modeled	multi product soybean	Iran
99	Rahbari, Arshadi Khamseh, and Mohammadi (2023)	strategic analysis for AFSC under crisis	develop a model of food supply chain under uncertain conditions in order to analyze it strategically during the pandemic, optimize production, storage, transportation, resources and exports in the supply chain	SSFSC	MILP	the export of canned food to neighboring countries with economic justification" was the best strategy for the company under study during the COVID-19 pandemic. this strategy reduce b supply chain costs and increase the human resources employed. Finally, the utilization of available vehicle capacity and the utilization of available production are optimal when using this strategy	SWOT method, PORTER and PESTEL, External Factor Evaluation (EFE) Matrix, the Internal Factor Evaluation (IFE) Matrix+ MILP +TOPSIS technique+ SEM,	generalization, one company in the study	canned food	Iran
100	Tan et al. (2023)	impacts of the pandemic on the AFSC	identify the challenges of AFSC, effective alternative strategies to be implemented by the government	AFSC	desk review	working under strict regulations came with a huge price paid for agrifood industries. This pandemic has affected the product availability and accessibility. Therefore, the government should be ready with the roadmap and enforce the measures to manage emergencies without disrupting the AFSC	literature review	limited geographic scope and time scale	NA	Malaysia

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101	Dong et al. (2023)	impact of digitalization on AFSC	investigates the relationships between digitalization and supply chain elements, particularly integration, communication, operation, and distribution, and their effects on corporate profitability and sustainability	IOAFSC	quantitative approach	synergistic effect of digital advancements leads to increased business profitability and sustainability	empirical investigation, PLS-SEM, statistical approach	study based only on a quantitative design	NA	Pakistan
102	Belamkar et al. (2023)	optimization of AFSC networking problem	develop a multi-objective model (multi-objective MILP model). Aims: maximize transportation profit, reducing transportation cost, reducing carbon dioxide emissions	PDL	multi-objective optimization problem (MOOP)	proposed model help to achieve balanced solutions between profit, cost and emissions	Weighted sum method (WSM), Fuzzy goal programming (FGP), Interactive Fuzzy satisfying technique (IFST), Global criterion method (GCM), and Weighted Tchebychef metrics programming (WTMP), study case	generalization, single period	apple	India
103	Azab et al. (2023)	modeling of sustainable AFSC with product perishability and environmental considerations	propose a model to maximize total supply chain profit and minimize total CO2 emissions	processing AFSC	mathematical approach	selection of plots to be planted, selection of the harvesting period, calculation of the transported quantity, determination of the processed quantity of products, estimation of the total stored quantity, determination of the satisfied demand and lost sales	multi-objective mixed-integer linear programming model MILP, study case, Pareto frontier	the model only optimizes economic and environmental dimensions	sugar beet	Netherlands
104	G. Wang, S. Li, Z. Zhang, et al. (2023)	visual knowledge map analysis	identify research hotspots and development trends, clarify the historical context, and explore future trends	CBAFSC	bibliometric analysis	since 2017, research in this field has shown a strong development trend, the resilience of the global food supply system have become the main lines of research in this field, research hotspots included cross-border agricultural product safety, cross-border agricultural supply chain systematization, and technology integration	PRISMA	sample literature from 2007 to 2022	NA	NA

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105	Karakoc et al. (2023)	Structural choke points	identify the spatially structural choke points	AFSC	complex network approach	the 7 structural choke points identified are generally consistent through time, particularly for processed food commodities	network statistics	perimeter	live animal and fish, cereal grains, agricultural products, animal feed, eggs, honey and other products of animal origin, meat, poultry, fish, seafood and their preparations, milled grain products and preparations, and bakery products, other prepared foodstuffs, fats and oils	United States
106	W. Yang, Xie, and Ma (2023)	the initial real information sharing in the AFSC	develops an evolutionary game model of AFSC participants +the impacts of the key parameters on the dynamic evolution process of participants	AFSC	modeling, simulation	the greater the risk coefficient of using blockchain technology, the higher the information cost of sharing initial real information, When the reward incentive coefficient, synergy coefficient, default penalty amount, and amount of sharing initial real information are larger, the participants are more inclined to keep the agreement. They abide by the rules of the blockchain system, actively share the initial true information, and strengthen the sharing of the initial true information through the incentive and punishment mechanism.	simulation experiments+ sensitivity analysis +the game payoff matrix of the two-party evolutionary game, Jacobian matrix and the sign of the determinant and trace	some factors affecting the sharing of the initial real information of the AFSC in the blockchain did not considered in the study	NA	China

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107	Melesse et al. (2023)	digital Twins in AFSC	*identify the benefits, types, integration levels, key elements, implementation steps, and challenges of DT in AFSC	AFSC	systematic literature review	DTs can improve efficiency and sustainability in the AFSC by providing visibility, minimizing bottlenecks, planning for contingencies, and improving existing processes and resources	descriptive and content wise analysis	document availability and coverage	NA	NA
108	Martella et al. (2023)	ecological Balance of AFSC	assess the environmental sustainability of AFSC	ISC	empirical approach	overall un-sustainability of the entire supply chain studied	case study , ecological footprint method	lack of detailed data	industrial tomato	Italia
109	Assis, Tümpel, et al. (2023)	Informality in AFSC	factors promoting informality	LAFSC	qualitative research	Factors that favor informality: family labor and temporary jobs, low education levels, lack of adequate supervision, cultural aspects, or other issues related to necessity and opportunity	interviews, case study, Descending Hierarchical Classification (DHC)	generalization	molluscs (mussels,oysters and scallops)	Brazil
110	Yontar (2023)	blockchain and circular economy of AFSC	research, analyze and prioritize the critical success factors of the use of blockchain technology	CLAFSC	multi criteria approach	blockchain contribute to the circular economy due to “ability to prevent food waste”, “Increased food security”, “Product lifecycle tracking” factors	survery, PESTEL analysis approach, analytic Network Process (ANP), Multi-Atributive Ideal-Real Comparative Analysis (MAIRCA)	limited expert sample	NA	Turkey
111	Derakhti, Santibanez Gonzalez, and Mardani (2023)	Challenges and barriers facing the AFSC	identify the challenges and barriers against the application of industry 4.0	AFSC	review of the Literature (SLR)	15 challenges classified into five major themes: technical, operational, financial, social and infrastructure. The four most important challenges identified are technological architecture, security and privacy, big data management and IoT (internet)-based infrastructure	systematically analyze the literature	restricted shade of selected items (24)	NA	NA
112	Gholian-Jouybari, Hashemi-Amiri, et al. (2023)	design of a sustainable AFSC network	simultaneously optimize:total profit, customer satisfaction, production level, water consumption.	AFSC	metaheuristic, convex robust optimization	water consumption is minimal, and a customer satisfaction level of 94.73%.	stochastic multiobjective programming model, case study, LP-metric method, hardness of the problems, Keshtel Algorithm, statistical comparison, multi criteria decision-making (MCDM)	generalization	saffron	Iran

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113	S. Yadav, Choi, et al. (2023)	IoT in AFSC	examine studies on IoT-based AFSC	AFSC	literature review (network-based cluster analysis	seven identified clusters, with the role of IoT (1:agri-food safety, traceability and sustainability; 2: AFSC sustainability; 3: AFSC performance measurement; 4: AFSC resilience in disruption; 5: AFSC integration and traceability; 6: AFSC transparency and coordination, and 7 identifies the barriers in IoT adoption	descriptive statistics and network analysis	limited period	NA	NA
114	Dovbischuk (2023)	logistics service quality	identify attributes of logistics service quality (LSQ) are associated with the superior LSQ	RAFSC, UAFSC	expectancy– dis confirmation paradigm in service management research and the related service quality (SERVQUAL) approach	service quality in agricultural logistics is a five-dimensional construct. Its five dimensions are reliability, digital transformation, corporate image, environmental sustainability, and quality of customer focus.	interviews, exploratory factor analysis (EFA)	data collection interrupted by war, generalization, exploratory only approach	NA	Ukraine
115	Peterson et al. (2023)	impacts of COVID-19 on AFSC businesses	identify changes in production, distribution and consumption following the pandemic	AFSC	statistical approach	impact varies depending on the regions and segments studied	regional survey, Wilcoxon rank-sum test	results limited to three regions	NA	US
116	Aris et al. (2023)	impact of technology 4.0 in business performance and sustainability of AFSC	identify the relationship between the adoption of agricultural revolution 4.0 towards AFSC business performances and AFSC sustainable agricultural performance encompassing economic, environmental and social dimensions in sustainability	AFSC	literature review	adopting AR 4.0 improves business performance (costs, quality, flexibility and speed) and the sustainability of the SC	conceptual study	conceptual study, without empirical validation	NA	Malaysia
117	Kopishynska et al. (2023)	application of modern enterprise resource planning systems	explores the possibilities of creating a unified digital information space in a modern cloud based enterprise resource planning (ERP) system to improve the management of all subjects in the territorial community and facilitate the transition to the Industry 4.0 technology landscape	RAFSC	quantitative and qualitative analysis	it is necessary to form a single software ecosystem on an ERP platform that is more flexible to support development	data collection, case study, business process analysis	lack of a secure digital system	NA	Ukraine

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118	**MontanyÃă117**	resilience factors of AFSC	extract those factors that have the greatest influence on the degree of resilience of the AFSC	AFSC	qualitative review of THE literature	there are at least three business management strategies, most frequently cited in the scientific literature, that have a positive effect on the resilience of the AFSC. They are: a customer orientation, local distribution and cooperation among agents	integrative review of publications	specific context of the COVID-19 pandemic	multi-product	multi-country
119	Babu and Devarajan (2023)	blockchain and Inter-Planetary File System (IPFS) for AFSC traceability	assessed a system for tracking agricultural products using blockchain's tracing and anti-tampering features	NPAFSC	exploratory approach	the proposed system (Blockchain, IPFS with encryption) improves decentralization and reliability compared to centralized approaches and other classic blockchains	simulation, experimental validation	limited constraints (non-perishable product)	NA	NA
120	Baležentis et al. (2023)	viable agri-food supply chains	define Methods and indicators used in the Analysis of the viability of the agricultural supply chain	AFSC	literature review	viability is not based only on short-term orientations and a return to the pre-crisis situation, but also on a long-term evolution towards the "new normal."	art analysis	lack of empirical data, lack of stakeholder integration	NA	NA
121	Ali, Sadiddin, and Cattaneo (2023)	risk and resilience in SMEs	define critical risks and strategies to build resilience in SMEs	AFSC	exploratory research	difference in risk and supply chain resilience profiles between developing and developed countries (adoption of digital technologies, training and development and support by national industry bodies)	interviews, cross comparative analysis	qualitative analysis only	NA	Pakistan, Tanzania, Australia
122	Ali and Govindan (2023)	digital transformation in AFSC	study impact of digital transformation in Operational Risk Extenuation (supply-demand mismatch, financial and transportation)	AFSC	empirical approach	extenuation of I4Ts help mitigate the impact of critical SC risks on firm performance	survey, statistical analysis, Structural equation model (SEM)	sample	NA	Australia
123	Philip and Marathe (2022)	impact of yield information sharing in AFSC	develop a new eco-label to promote an efficient agricultural method that provides the right balance between yield and environmental impacts through the optimal use of these farm inputs	PAFSC	analytical approach	this model allows to optimize the farm inputs, retail prices, and order quantity so that the stakeholders' profits are maximized, stakeholders must share yield information when environmentally conscious consumers dominate the market	game-theoretic model	two echelon, theoretical model, very simplified chain with one farmer, one retailer, one producer	NA	NA
124	Tuni and Rentizelas (2022)	improving environmental sustainability	showcase how assessing the environmental sustainability performance of a multi-tier food SC made up by SMEs can support decisions in order to drive evidence-based green improvements in the SC operations	GAFSC	empirical approach	applying an eco-intensity method to a food chain composed of SMEs allows identifying hotspots and reducing overall environmental impact, while facilitating the implementation of multi-level GSCM over time	eco-intensity method, longitudinal case study	micro enterprises, limited geographical scope	bread (Patto della Farina)	Italia

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
125	Ivo de Carvalho, Relvas, and Barbosa-Póvoa (2022)	roadmap for performance assessment	define a standardized mapping of AFSC to be considered in a PMS, identifying the key stakeholders, material, information, and financial' direct and reverse flows, identify the most critical sustainability-related Performance Areas (PA) (categories) that should be measured in any AF-SSC, and for each PA proposed, detail the main subcategories (characteristics)	GAFSC	analytic approach	the proposed framework to guide the sustainable measurement of different stakeholders CONTAINS, that includes 16 performance areas and 71 subcategories	performance Measurement Systems (PMS), Triple Bottom Line (TBL), interviews, literature review	until October 2021, only two concepts studied (CE, sustainability)	NA	NA
126	Z. Li, P. Zhao, and X. Han (2022)	recovery of AFSC network disruption	analyze disruption propagation and recovery in AFSC by introducing weak ties and evaluating strategies to improve resilience	RAFSC	empirical and case study	combining strengthening existing relationships with establishing new ones enhances resilience	optimization and statistical modeling	generalization	NA	China
127	Stillitano et al. (2022)	multi-cycle model for measuring the sustainability	propose a multi-cycle approach with expanded assessment boundaries, including co-products, in an attempt to internalize circularity impacts	circular AFSC	mixed-methods approach	customized LC framework proposed can better highlight the environmental and socioeconomic performances of the system of cycles, allowing CE to deliver its promises of sustainability	Life Cycle Assessment (LCA), Environmental Life Cycle Costing (ELCC), Social Life Cycle Assessment (SLCA) in terms of Psychosocial Risk Factor (PRF) impact pathway	exclusion of distribution, use, and disposal in base LCA	olive oil	Italy
128	Anastasiadis et al. (2022)	role of traceability	explore the role of traceability in the transition to sustainability driven AFSC	circular AFSC	mixed-methods approach	change at the consumer level drives changes in the supply chain and eventually at the company level	survey , Delphi rounds, Categorical regression	sample of consumer survey	tomato	Greece
129	Ada (2022)	Sustainable supplier selection in AFSC	identify the criteria for sustainable supplier selection	DAFSC	mixed-methods approach	performance assessment must rely on the following criteria: material cost, transportation cost, quality assurance, innovation, energy saving, green manufacturing, communication skills, customer satisfaction, productivity, social responsibility, supplier reputation, information technology systems and environmentally friendly technology	literature review, Fuzzy Analytic Network Process (FANP), fuzzy VIKOR	sample (seven agri-product supplier companies)	NA	Turkey

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
130	P. Kumar and Rajesh Kumar Singh (2022)	strategic framework for developing resilience	identify the major impacts of COVID-19 on AFSC and their correlation with different remedial approaches/strategies to improve the resilience of the AFSC	AFSC	mixed-methods approach	supply chain collaboration is the most important strategy, coordination between the stakeholders, information sharing	Quality Function Deployment (QFD) approach, Best-Worst method, Application of Quality Function Deployment (QFD)	limited sample	food grains (perishable items)	India
131	Tirkolaee et al. (2021)	decision-making approach for green supplier selection	integrate MCDM and robust optimization to determinate the optimum order quantity of each supplier under uncertain conditions	PAFSC	mixed-methods approach	model validation	AHP-fuzzy TOPSIS, RGP approach, multi-objective mixed-integer linear programming, robust goal programming	method based only under economic and GSS criteria, generalization	meat, chicken, schnitzel, rice, fish, tomato paste, lemon juice, tomato, and yogurt	Iran
132	Boenzi et al. (2022)	environmental impact of fresh products compared to semi-finished products (processing of fresh products by mechanical and thermal methods) in the production process	evaluate the environmental impact of each step in AFSC	PHL	empirical approach and life cycle assessment	using fresh products has a lower environmental impact than using semi-finished products, because the perishable nature of the products limits the transport distance of raw materials	impact assessment method, ReCiPe method	case study, seasonality of products and inter-annual variability of data	Apricot jam	Italy and Europe

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
133	Assis, Lucas, and Rainho (2022)	trust in AFSC	carry out a meta-analysis on the relationships of trust among the stakeholders in agrifood supply chains	AFSC	analysis approach	trust remains a central theme	meta-analysis, bibliometric study	limited perimeter: “Agricultural and Biological Sciences” area, "Food Science Technology” , “Agricultural Economics Policy” , “Agriculture Multidisciplinary” , “Agronomy” , “Fisheries”, “Agriculture Dairy Animal Science” , and “Horticulture”	NA	NA
134	Navarro-del Aguila and Burgos-Jiménez (2022)	developing hybrid AFSC	identify the stages and the tasks of this short chain, compare them with conventional agri-food chains, approve the hybrid model of AFSC	SASC, PAFSC	emprical approach	contribution of the hybrid model: the sharing of personnel, infrastructures and services, complementarity in the product range, thus avoiding food waste, knowledge of consumer tastes and needs	analysi, interviews, study case analysis	generalization, time-limited assessment (only short term)	fresh fruit and vegetables((peppers, tomatoes, courgettes, cucumbers, aubergines, green beans, watermelons, and melons)	Spain
135	Prajapati et al. (2022)	circular economy in AFSC	minimize collection costs	AFECC	mathematical approach	innovative packaging materials can help to minimize the wastage of food, the cost of collection varies with an increase in the number of nodes	Mixed-Integer Non-Linear Programming (MINLP)	no multi-echelon durability, simulated data	grains	India
136	Agnusdei and Coluccia (2022)	research trends	identify research areas in the field of sustainable AFSC and the role of technological innovation in the transition to a sustainable AFSC	circular AFSC	literature review	the need for the transition to sustainable AFSC is developing rapidly, digital technologies are scarcely studied within the AFSC sustainability field, post-consumption phase is a challenging issue	Bibliometric, analysis, network analysis and content analysis	transition to a holistic and multidimensional approach	NA	NA
137	Takavakoglou, Pana, and Skalkos (2022)	constructed Wetlands in AFSC	identify the role of constructed wetlands as nature-based solutions in the AFSC of the forthcoming post-COVID-19 era	AFSC	literature review	numerous applications for constructed wetlands at every stage of AFSC, contribution of constructed wetlands to the strategic goals of the Horizon Europe Strategic Plan	emergent qualitative analysis (deductive and inductive)	generalization	NA	Europe

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138	Kontopanou and Tsoulfas (2022)	triple "A" AFSC model	investigate the impact of agility, adaptability, alignment in AFSC	AFSC	literature review	The triple "A" models promote the increase of the AFSC effectiveness	literature review, analysis and comparison	selective bibliography study	NA	NA
139	Ali, Gölgeci, and Arslan (2023)	knowledge management practices, risk management culture	investigate the mechanisms of exposure to supply chain risks	small AFSC	empirical cases	Knowledge application fosters RMC (Risk Management Culture), which enhances SC resilience	quantitative cross-sectional survey, SEM for data analysis	single country	NA	Australia
140	S. Yadav, Luthra, and Garg (2022)	IoT-based coordination system	identify enablers, develop a framework, analyze causal relationships	AFSC	MCDM approach	'Top Management Support (TMS)' is the main driver for improving coordination	ISM- Fuzzy DEMATEL	expert sample	NA	India
141	Meemken et al. (2021)	impacts of sustainability standards	review the existing evidence on sustainability standards	GAFSC	Narrative review methodology	Standards can improve some production practices but are inadequate for achieving widespread food system sustainability and do not significantly advance equity	SLR	limited by the scope, quality, and methodological constraints	products governed by sustainability standards (e.g., coffee, fruits, vegetables)	NA
142	Joshi, Rohit Kumar Singh, and M. Sharma (2020)	AFSC practices	identify the factors influencing the adoption of sustainable agribusiness practices	AFSC	case study	Main barriers: low adoption capacities and absence of cohesive sustainable policy.	semi-structured interviews	sample	NA	India
143	Kusumowardani et al. (2022a)	Food loss and food waste (FLW)	develop a circular capability framework	AFSC	case study	circular economy adoption is limited; preventative strategies are lacking.	interviews	sample	NA	Indonesia
144	S. Yadav, Luthra, and Garg (2021)	IoT-driven multi-tier system	analyze cause-effect relationships for an IoT	AFSC	MCDM approach	hierarchical and causal model of enablers	ISM- Fuzzy DEMATEL	expert sample	NA	India
145	Sisto (2025)	blockchain in AFSC	Examine impact of blockchain technology for traceability, sustainability, and certification in AFSC	AFSC	conceptual approach	blockchain technology can improve supply chain management, profitability, and transparent relationships	analysis of regulations	Technology cannot replace totally certification processes	NA	EU/USA
146	V. S. Yadav et al. (2022)	current status and challenges of AFSC and key indicators to measure the performance of AFSC	identify the various challenges in AFSC, review the research contribution in the field of designing AFSC network (AFSCN), investigate performance measurement system of AFSC through various performance indicators	AFSC	literature review	technology adoption and integration in design AFSCN are exciting prospects for future AFSC research	Qualitative content analysis	consideration only of the concept of sustainability	NA	NA

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
147	Mehmood et al. (2021)	drivers and barriers for a circular economy	identify the drivers and barriers for implementing circular economy initiatives	AFSC	literature review	environmental, policy and economy , and financial benefits are the three top drivers. Institutional , financial , and technological risks are the top three barriers	Data analysis	subjectivity in methodology(keywords, inclusion and exclusion criteria)	NA	NA
148	Amentae and Gebresenbet (2021)	digitalization in AFSCS	identify the potential contribution of digitalization of the food system	AFSC	literature review	'Blockchain' was the top hit among keywords, the challenges are infrastructure and cost, knowledge and skill, law and regulations, the nature of the technologies, and the nature of the food system	data analysis	time scale	NA	NA
149	A. Rejeb, K. Rejeb, and Zailani (2021)	big data in AFSC	identify the contribution of big data to the sustainable development of AFSC	literature review	big data help in crop/plant management, animal management, waste management and traceability	PRISMA	lack of empirical validation	NA	NA	
150	Chiaraluce, Bentivoglio, and Finco (2021)	circular economy in AFSC	identify the trends and future pathways	AFSC	literature review	the topic CE in AFSC is of particular relevance in the scientific community, and the concept CE is continuously evolving	bibliometric analysis	horizon ends at 2019 and publication at 2021	NA	NA
151	Barbosa (2021)	research streams on AFSC	identify frequently-used keywords, new and hot research topics and frequently-studied supply chain management practices	AFSC	literature review	frequently used keywords are food supply chain, food waste, sustainability, food safety, SCM, food industry, and food security. New research themes are contract, blockchain, internet of things, resilience, and short food supply chain	Bibliometric analysis	scope limited to the Web of Science (WoS) database only	NA	NA
152	Galera-Quiles et al. (2021)	eco-Innovations and exports interrelationship	analyze the influence between the two variables, both at a business (micro level) and at a country/region (macro level), and identify the most influential factors (country of origin and sector of activity)	IAFSC	literature review	there is a positive bidirectional relationship, influenced by several factors, such as social performance, environmental regulation, cooperation strategies, employment level, or business size	systematic literature review	Methodological limitation(bibliometric approach)	NA	NA

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
153	Santos, Torrisi, and Pantoni (2021)	third party certification of AFSC using smart contracts and blockchain tokens	demonstrate a harvest-level third-party certification (TPC) mechanism for agri-food using Ethereum smart contracts and ERC-1155 tokens to publish tamper-resistant certificates linked to harvests, enabling public access via mobile apps and reducing fraud/greenwashing along the supply chain. The approach aims to track origin and quantity of each harvest from farm to consumer, incentivize chain participants, and provide immediate certificate access through a URI on the authority's site	AFSC	mixed approach	harvest-level TPC via ERC-1155 NFTs with cert links, proven in a PoC on Ethereum to enhance transparency and reduce fraud.	Harvest-level Third Party Certification (TPC) using ERC-1155 NFTs to mint harvest-specific tokens with cert URIs; tokens flow along the supply chain and can be read by consumers via a mobile app; PoC on Ethereum (Rinkeby/mainnet) shows tamper-resistant, transparent provenance and incentives for participant	generalization	plant agricultural products (e.g., wine grapes)	NA
154	Hajimirzajan et al. (2021)	large-scale crop planning	minimize total supply chain costs and guarantee farmers' minimum income using NGA/RCG coordination	UAFC	mathematical modeling	a large-scale, climate-aware crop planning framework that minimizes total supply chain costs	generalization	potatoes, onions, tomatoes	Iran	
155	F. Taghikhah et al. (2021)	extended AFSC	model and analyze an extended sustainable SC by integrating production decisions with consumer behavior, capturing heterogeneity in demand, and evaluating social, environmental, economic, and behavioral performance	organic and conventional AFSC	mixed approach	including consumer heterogeneity and behavioral dynamics in an extended agro-food supply chain improves understanding of sustainable transitions	agent-based modeling (ABM), discrete event simulation (DES), and system dynamics (SD)	limited empirical data	wine	Australia
156	Rao, Bast, and Boer (2021)	european private food safety standards	assess how European private food safety standards interact with global AFSC	GAFSC	literature review	the most studied European private food safety standards are owned by retail conglomerates and therefore place the retail sector in a position of influence in the supply chain. These standards influence supply chain structures, market access, and the efficiency of food safety management systems	PRISMA	European focus	NA	EU

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157	Bottani et al. (2025)	key performance indicators	determine the map, analyze, and classify KPIs in the AFSC	AFSC	literature review	KPIs for the Food Supply Chain are systematically reviewed and mapped	bibliometric analysis, content analysis	KPIs are derived from only 125 papers accessed on 31 December 2024 and may not be exhaustive	nine product types (Agricultural products, Meat, Fish, Dairy products, Bread, Animal and plant production, Alcohol-free drinks, Food production, Fruit)	NA
158	Yu et al. (2025)	empirical transformation	identified technology impact in AFSC to map themes and theories; identify trends and future directions	AFSC	literature review	five main clusters: behavior intention, sustainability, operation, performance, and risk management. importance of integrating digital technologies to enhance efficiency, transparency, and sustainability in AFSCs	PRISMA, structural equation modeling	limited scope	NA	NA
159	M. Kumar et al. (2022)	sustainable consumption and production	identify circular-based sustainable KPIs for AFSC to achieve sustainable consumption and production, rank KPIs and select best circular alternatives	Circular AFSC	MCDM approach	Economic dimension has highest weight	F-AHP , F-TOPSIS	generalization	dairy products	India
160	Manikas et al. (2022)	food security	develop a conceptual framework to assess AFSC and identify the main global drivers that affect the local food systems	AFSC	conceptual approach	FS indicators to support an early warning system and policy decisions for resilient UAE AFSC	delphi groups, case studies	generalization	NA	UAE
161	Kusumowardani et al. (2022b)	circular capability	develop a circular capability framework	UAFC	empirical and case study	proposed framework can enhance the reduction FLW, improve food security, price stability, economic resilience, and environmental benefits	semi-structured interviews , observations	lack of validation	fruits and vegetables	Indonesia
162	Bager, C. Singh, and Persson (2022)	blockchain in AFSC	assess the potential of blockchain technology to promote sustainability in supply chains through increased traceability and transparency and to identify barriers	IAFC	empirical and case study	digitalization of supply chain increase efficiency and reduce costs, disputes, and fraud, while providing more insight end-to-end through product provenance and chain-of-custody information	semi-structured interviews	generalization	Coffee	Colombia and Denmark

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
163	Giovannetti, Bertolini, and Russo (2021)	role of cooperation	Assess how Commons and Social Capital through cooperation enhance sustainability and resilience in supply chain	UAFC	empirical and case study	Cooperation and open, rights-based SC improve transparency, pricing fairness, and sustainability	depth case study, interviews	Single case	Parmigiano Reggiano cheese	Italy
164	Beber, Langer, and Meyer (2021)	strategic actions for internationalization	investigates strategies for integrating supply chains and how both sides can benefit from this situation also in terms of sustainability	AFSC	empirical and case study	viable sustainable internationalization strategies with emphasis on governance, market entry , differentiation, R&D training, quality improvements, and supportive policy for win–win outcomes	expert interviews, content analysis, snowball sampling method	generalization	dairy products	Brazil, Germany
165	Kramer, Bitsch, and Hanf (2021)	blockchain in AFSC	determine the impact of the three types of blockchain platform in AFSC governance and coordination	AFSC	empirical and case study	blockchain enhance coordination, information sharing, decision-making, and collective learning benefits	expert discussions	lack more empirical validations	coffee and frozen fish	EU
166	**PÃľrez165**	collaborative relationships	analyze collaboration patterns to improve sustainability and performance	PAFSC	empirical and case study	collaboration is more profitable than collaboration with customers	survey, regression	single regional chain	NA	Spain
167	Ciccullo et al. (2021b)	role of food waste prevention technologies	investigates extensively the range of the available technologies and the detailed objectives of such technologies for food loss and waste prevention	UAFC	empirical and case study	Collaboration with technology providers can be based on continuous technical assistance and consulting for data elaboration and data analysis as well as on full data sharing and co-design, allowing to achieve a different impact on food loss and waste prevention	case analysis	limited sample	fruits and vegetables	Italy
168	Zaridis, Vlachos, and Bourlakis (2021)	strategy and scale constraints	examine the impact of strategy and scale constraints on AFSC collaboration	AFSC	empirical and case study	collaboration positively affects small and medium-sized enterprises performance	principal components analysis, hierarchical regression, cluster analysis, correlations	limited sample	NA	Greece
169	Yu Cao et al. (2020)	impact of the green standards at different stages of AFSC	design a cost sharing contract and a buyback contract to coordinate the greening effort decisions	GAFSC	mathematical modeling, theoretical analysis, and numerical simulation	increasing green standards can stimulate the supply chain to increase greening efforts and sales price, which increases costs and decreases product quantities, consequently decreasing the supply chain profits. The greening effort investment by one entity of the supply chain could be affected not only by its own standard, but also by the standard for the other entity	game model (Cobb–Douglas and demand function)	generic (no distinction between products type)	NA	NA

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170	P. Liu, Long, et al. (2020)	impact of blockchain and big data in coordination of GAFSC	analyze the benefit models of producer and retailer + propose a cost-sharing and revenue-sharing contract to coordinate the AFSCS	GAFSC	mathematical modeling, simulation	when the total investment cost payed by producer and retailer is in a certain range, using ISBD will help chain members gain more benefits. If chain members want to gain more benefits after using ISBD, they should try their best to optimize costs by extracting valuable information	Stackelberg game theory	SC with one producer and one retailer	NA	China
171	Nattassha et al. (2020)	circular economy in AFSC	develops a conceptual model of an organic fertilizer producer	GAFSC	soft systems methodology approach	the importance of increasing interaction between the company and the farmers	CATWOE framework	one echelon	organic fertilizer derived from vegetables and fruit	Indonesia
172	Mehdiyeva et al. (2020)	barriers and Drivers of the Implementation and Management of GAFSC	analyze the relationship between logistics investments (development of transport and storage capacities) and agricultural performance	GAFSC	quantitative approach	the major driver of the implementation of green supply chain management is the demand for high-quality products and the availability of modern infrastructure.	statistical analysis, documentary analysis	limited data	NA	Azerbaijan
173	C. Han et al. (2020)	home-delivery-oriented agri-food supply chain (HASC) alliance AND THE competitiveness OF small and medium agri-food enterprises	propose a theoretical framework for the HASC alliance and its implementation strategies	LMAFSC	analytic method	the minimum and maximum cost ranges of control strategies at which the alliance can maintain its stability and performance are 5% and 29%, respectively, of the total operation cost	literature review, simulation	generalization	NA	China
174	Cui, Guo, and H. Zhang (2020)	coordinating a green AFSC with revenue-sharing contracts	explore the impact of revenue-sharing contract	GAFSC	mathematical modeling	the revenue-sharing contract is beneficial to increase the greening level, improve both the farmer's profit and the retailer's profit. The effectiveness of the revenue-sharing contract is positively correlated with consumers' sensitivity to the greening level	Game theory, numerical simulation	single farmer and single retailer	NA	NA
175	Schrobback, Rolfe, and Star (2020)	AFSC assessment	develop a supply chain assessment framework	UAFC	empirical and case study	description matrix and framework	literature review, description and analysis AFSC	limited time series	beef and grain	Australia
176	Hao Fu et al. (2020)	blockchain in AFSC	study the coupling of management mechanisms between blockchain and the AFSC in the "technology, institution" logic	AFSC	descriptive case studies	blockchain reduces information asymmetry, limits opportunism and is built around organizational structure, trust and contract.	study case	case-based study	poultry, grain	China

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177	Kamble, Gunasekaran, and Rohit Sharma (2020)	data-driven AFSC	propose an application framework for practitioners	AFSC	literature review	Environmental sustainability has the most attention	SLR	2000-2017	NA	NA
178	Pancino et al. (2019)	partnering and the sustainability of AFSC	design of the process of designing a multi-stakeholder partnership to be implemented in the agri-food supply chain in order to introduce sustainable agricultural practices	MTAFSC	Empirical	it is not possible to design a single standard contract that also includes variation for different locations or division of output between the stakeholders	case study, empirical analysis (documentation, interviews, and participant observations)	generalization	Barilla products (tomato, durum wheat, and oilseed crops)	Italy
179	Jonkman, Barbosa-Póvoa, and Bloemhof (2019)	design AFSC	integration of seasonality and harvesting decisions, perishability and processing into the design of the AFSC	PAFSC	mathematical modelling	maximize the total gross margin and minimize the global warming potential in CO2 -eq	pareto-efficient frontier, ε-constraint method	large differences between time intervals of expected harvest yield	sugar beet	Netherlands
180	Yuliang Cao and Mohiuddin (2019)	price formation mechanisms	analyze the price formation mechanisms	upstream and downstream AFSC	empirical and case study	Price forms at wholesale node and cascades to upstream farmers and downstream retailers/consumers	descriptive statistics	small sample	fresh and raw vegetables (cucumbers)	China
181	Banasik et al. (2019)	accounting for uncertainty	propose a multi-objective two-stage stochastic programming model to analyze and evaluate the economic and environmental impacts to account for uncertainty in AFSC	UAFC	mixed approach	applying a 2-stage stochastic programming approach for production planning decisions can lead to improved economic and environmental performance in an AFSC	stochastic programming, deterministic model	uncertainty limited to yields and demand, single case study	mushrooms	Netherlands
182	Hongyong Fu et al. (2018)	Weather risk–reward contract	design a weather risk-reward contract mechanism	AFSC	mathematical approach	this contract can settle the distortion of the sustainable investment level and effectively motivate farmers to participate in the sustainable agricultural practice	Guaranteed Price Mechanism	lack of social pillar	NA	China

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
183	Mangla et al. (2018)	enablers to implement sustainable initiatives	analyze the key enablers in implementing sustainable initiatives for AFSC	retail supply chain	empirical case study	three bottom level enablers HAVE the highest independence powers. These enablers are influential enablers. Six enablers were categorized in the cause group and the remaining four in the effect group	literature review, expert consultation, Interpretive Structural Modeling (ISM) - fuzzy Decision Making Trial and Evaluation Laboratory (DEMATEL), MICMAC analysis	the subjective cause, effecT diagram build on the judgments of experts from a particular industry. Further, the identification of the enablers could be challenging	fresh produce (vegetable and fruits), dairy products, and fast-moving consumer goods (FMCG)	India
184	Huilin Chen et al. (2021)	blockchain and Deep Reinforcement Learning in AFSC	design a blockchain-based ASC framework, propose a Deep Reinforcement learning based Supply Chain Management (DR-SCM) method	AFPSC, RSC	experimental, simulation	blockchain ensures reliable product traceability, DR can achieve higher product products than heuristic and Q-learning methods.	simulation, RL algorithm	experiences limited to simulated scenarios	corn	NA
185	Stone and Rahimifard (2018)	resilience in AFSC	identify the aspects of resilience	AFSC	literature review	complexity of AFSCs and subsequent exposure to almost constant external interference means that disruptions cannot be seen as a one-off event	SLR	one aspect	NA	NA
186	Dania, Xing, and Amer (2018)	collaboration in AFSC	investigate the landscape of extant literature	AFSC	systematic review	10 key behavioral factors to enable an effective collaboration system for sustainable AFSC :joint Efforts, sharing activities, collaboration value, adaptation, trust, commitment, power, continuous improvement, coordination and stability	resource Dependence Theory (RDT), Theory and Content Analysis , analysis based on frequency	too generic factors, little-studied interactions	NA	NA
187	Luo et al. (2018)	bibliometric and content analyses	identify the homogeneous areas in AFSC and to assess the movement and interactions within and between fields	AFSC	bibliometric and content analyses	6 clusters are identified : short/alternative food supply chains, ASC Sustainability, modeling ASC traceability, risk management, and optimization, global ASCs, ASC transparency and traceability, ASC relationships/vertical coordination/networks	systematic literature review combined with bibliometric and content analyses	subjectivity of article selection	NA	NA
188	Pappa, Iliopoulos, and Massouras (2018)	electronic traceability systems (ETsystems)	investigate the factors influencing the acceptance and use of ETsystems in AFSC	GAFC	mixed approach	'Perceived Control' and most importantly, the 'perceived costs' over the installation and operation of the ET-system, is the most important factor with the strongest direct effect influencing the intention to install and operate such a system	technology Acceptance Model (TAM2), theory of Planned Behavior (TPB), PLS-SEM analysis	generalization	dairy products	Greece

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
189	G. Zhao, S. Liu, and Lu (2018)	theory of AFSC resilience	identify AFSC resilience factors (enablers, model and their interrelationship) and develop a comprehensive resilience framework	AFSC	mixed approach	leadership and working-team stability are top driving factors; traceability and information sharing are critical but largely dependent on others	structural modeling total interpretative and cross-impact multiplication matrix	limited interview coverage	NA	United Kingdom, France, Spain, Italy, Argentina
190	Allaoui et al. (2018)	sustainable AFSC design	optimizing the three pillars of sustainability	AFSC	mixed approach	two-stage hybrid approach yields a Pareto frontier showing trade-offs between cost, emissions, water use, and jobs	AHP, OWA for partner efficiency, multi-objective MILP, Pareto front via ε-constraint	data limited to case study	NA	NA
191	Stranieri et al. (2021)	blockchain and AFSC performance	assess impact of blockchain on the AFSC performance	PAFSC	empirical and case study	blockchain improve information accessibility, availability and sharing. However, the flexibility and responsiveness of operations within the AFSC do not impacted	case study, thematic analysis	generalization, limited sample	Poultry meat, lemons, and oranges	Europe
192	Q. Hu, Q. Xu, and B. Xu (2019)	introducing of online channel in AFSC	investigate the practicality of introducing online channels in the AFSC	GAFSC	mathematical optimization	the green food brand effect of online channels is does not necessarily improve with scale, and the initial freshness has a positive relationship to the profit, demand, and price of farmer cooperatives and online retailers	game theory	constant price during product shelf life	NA	NA
193	C. Zhang et al. (2016)	bio-refinery technologies for unused biomass	impact of bio-refinery technologies for unused biomass in sustainability of AFSC	AFPSC	experimental approach	improving agricultural production efficiency is not enough to achieve sustainable global food security, developing bio-refinery technologies for unused biomass in the agri-food supply chain could be the crucial factor	technique practical experimental	laboratory tests	green tea residue	china
194	Garofalo et al. (2017)	environmental sustainability of AFSC	identify the impacts related to greenhouse gas emissions	AFSC	International Reference Life Cycle Data System (ILCD)	waste produced during the processing phase was the main contributor to the total environmental impact, followed by the packaging and cropping phase	life cycle assessment methodology	partial sectoral study	whole-peeled tomato	Italy
195	Tasca, Nessi, and Rigamonti (2017)	environmental sustainability of AFSC	study the combination of organic farming and alternative distribution (local and large-scale) and their impact to reduce the environmental footprint compared to the conventional model	RAFSC	empiric	no agricultural techniques have a better overall environmental profile, distribution of raw organic product loose is generally the most advantageous in the majority of impact categories	life cycle assessment LCA	limited perimeter	endive	Italy

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
196	Akhtar, Tse, et al. (2016)	adaptive leadership and sustainable AFSC	study the link between adaptive leadership and sustainability	IAFSC	Descriptive statistics	data-driven and adaptive leadership is a key determinant for sustainability	structural equation modeling(SEM)	survey research	dairy,meat,vegetables and fruits	from/to China,India and New Zealand and another country
197	Aggarwal and Srivastava (2016)	collaboration in AFSC	study the impact of collaboration on AFSC efficiency	AFSC	empirical approach, qualitative methodology	collaboration brings benefits to buyers and suppliers as well as the application of sustainable best practices in the agi-food industry (reduction of waste, price vocation, etc)	case Study, interview, grounded theory methodology	single case study, generalization	rize	India
198	Accorsi et al. (2016)	ecosystem carbon balance in AFSC	develop A linear programming model to optimize agriculture, and logistics costs and minimize the carbon emissions	AFSC	mathematical modeling	planning focused solely on profitability leads to imbalances in the sustainability of agri-food chains, hence the importance of integrating environmental criteria	linear programming (LP), study case	scale restriction	potato	Bologna
199	Galal and El-Kilany (2016)	uncertainty of demands in AFSC	study the effect of changing the order quantity and lead time in AFSC on economic and environmental performance measures (stock levels, associated costs and CO2 emissions)	PAFSC	analytical simulation approach	reducing order quantities can reduce costs and emissions; in addition, it is important to integrate the emissions because costs IS not sufficient for chain performance	discrete event simulation model, case study	data reliability (developing countries)	oranges	from Egypt to Netherlands
200	Akhtar and Z. Khan (2015)	leadership and coordination between AFSC ACTORS	impact of participative leadership and directive leadership in AFSC coordination	IAFSC	empirical approach	participative leadership approach is highly significantly associated with the effectiveness of AFSC coordination unlike the directive leadership	survey, covariance-based structural equation modelling (CBSEM), exploratory factor analysis, reliability, and validity tests	generalization	dairy, meat, apples, onions, and wine	New Zealand-UK
201	Rota, Zanasi, and Reynolds (2014)	AFSC governance structures and sustainability	determine the impact of processes oriented sustainability on the supply chain governance structures	AFSC	empirical approach	the integration of sustainability dimensions into the governance structures of agri-food chains, strengthening technical and logistical efficiency and the improvement of food quality	adoption of experts' evaluations	complexity, generalization	soy and beef	Latin America

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S/N	Author(s)year	Topics focus	Objective(s)	Scope	Methodology	Key observations	Research methods	Limitations	Product	Country
202	Folinas et al. (2014)	lean thinking practices in green AFSC	propose a systematic approach to determine waste (by measuring the time without added value in production and logistics processes, as well as the water and energy consumed)	GAFSC	systematic approach	integration of Lean techniques, VSM, and ERP modules improves the performance of the AFSC	study case, value stream mapping (VSM)	limited generalization	corn for animal feed	Greece
203	Tsolakis et al. (2014)	decision making in AFSC	recognize the hierarchy of the decision-making process for the design and management of AFSCs (Strategic, tactical and operational levels)	AFSC	literature review	decision-making processes in the agri-food supply chain are dynamic and stochastic in nature; need to develop real-time optimization tools to solve dynamic and stochastic problems in the AFSC	synthesis of the literature	static logistics chain	NA	EU
204	Verdouw et al. (2014)	TIC in AFSC	design a platform for smart agri-food logistic information systems	AFSC	user-oriented approach	TIC facilitates real time virtualization, logistics connectivity, and logistics intelligence of AFSC	study case (needs, specifications, technical design)	generalization	fresh fruit and vegetables (FFV) industry and the Plants and Flowers (PF) industry	EU
205	Boudahri et al. (2013)	configuration AFSC network	develop a clustering-based location-routing model	PSL	mathematical modeling approach	minimize the total costs of the AFSC, determine the location of the slaughterhouses, and determine the distribution routes to the customers	Branch and Bound	one echelon of AFSC	chicken and turkey	Algeria
206	Vorst, Kooten, and Luning (2011)	integration of product quality in TCAFSC	develop a preliminary diagnostic instrument, which can be used to assess supply chain networks on QCL(quality controlled logistics) and identify opportunities for improvement	TCAFSC	exploratory case study	using time-depend quality information helps to optimize product quality and availability in markets and minimize shrinkage	literature review, study case	limited sectoral scope	tomato	Netherlands
207	Fischer et al. (2009)	contractual choice and sustainable relationships	identify factors influencing the choice of contract type and which lead to sustainable inter-enterprise relationships	DTAFPSC	empirical approach	market, industry, and enterprise-specific characteristics influence the choice of contract type	survey, statistical data analysis (binary logit model + structural equation model SEM))	limited sectoral scope	pig meat, beef, and cereals	Germany, UK, Spain, Ireland, Finland, and Poland
208	Aramyan et al. (2007)	performance measurement	develop a generic framework and evaluate it	CBSC	conceptual framework, case study	identifies 4 main categories: efficiency, responsiveness, food quality, flexibility	conceptual framework analysis	absence of empirical validation and integration of multi-criteria indicators	tomato	Netherlands & Germany

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209	Rohit Sharma, Kamble, et al. (2020)	Machine learning (ML) applications in sustainable AFSC	Identify the current state and future potential of ML applications in AFSC	AFSC	literature review	Artificial neural networks (ANN) are the most frequently used ML algorithm	Systematic literature review + descriptive statistic	2000-2019	NA	NA
210	Nayal et al. (2023)	blockchain in AFSC	identify the factors that affect blockchain technology adoption in the AFSC	AFSC	empirical approach	blockchain technology adoption has a positive effect on sustainable supply chain performance	SEM	sample	fresh fruits, vegetables, beverage, dairy	India
211	S. Kumar et al. (2021)	circular economy adoption barriers	identify ranking of barriers	AFSC	Multi-criteria approach	Most critical barriers are lack of government support and incentives (LGS) and lack of effective policy and protocol (LPP)	Delphi, ISM, and ANP	Generalization	pulp, milk, jute , rice, sugar cane, wheat, cotton, vegetables, and fruit	India
212	Kamble, Gunasekaran, and Gawankar (2020)	Blockchain technology in AFSC	identify the enablers that encourage blockchain adoption in AFSC	AFSC	MCDM approach	Most significant enabler is traceability	Delphi, ISM-MICMAC	Generalization	NA	India

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